Australian Government
Department of Industry, Science and Resources

Research and Development Tax Incentive

R&D tax incentive application

Tracking ID:	PCYNEHCNX
Submitted:	30 Apr 2026 03:50 PM AEST
Company name:	CAPIX TREASURY SOFTWARE PTY LTD
Australian Business Number (ABN):	81072587976
Australian Company Number (ACN):	072587976
Registration Date:	24/01/1996
Income period:	01 Jul 2024 - 30 Jun 2025
Financial year:	2024-25

Table of contents

Registration type

For help with completing this section, please refer to the **Application Guidance**.

Is the company registered with the Australian Securities and Investments Commission?

- [x] Yes, under an Australian law
- [ ] Yes, under foreign law that is an Australian resident for tax purposes
- [ ] Yes, under a foreign law AND
is a resident of a country with which Australia has a double tax agreement, including a definition of 'permanent establishment' AND
is carrying on business in Australia through a permanent establishment as defined in the double tax agreement
- [ ] No, this company is not registered with Australian Securities and Investments Commission

Company details

For help with completing this section, please refer to the **Application Guidance**.

What date was the company registered with the Australian Securities and Investments Commission?

You can find this information in the Australian Securities and Investments Commission register at ASIC Connect. Please notify the Australian Securities and Investments Commission if your details need to be updated.

24/01/1996

Is the company the head of a consolidated or multiple entry consolidated group?

Only the head company of a consolidated or multiple entry consolidated group can apply to register R&D activities. The head company must register R&D activities performed by any member of the group. For further information on claiming the R&D Tax Incentive if you are a member of a consolidated or multiple entry consolidated group please visit the Australian Tax Office website.

- ○ Yes
- ☑ No, the company is not part of a consolidated or multiple entry consolidated group
- ○ No, the company is a subsidiary of a consolidated or multiple entry consolidated group

Is the company controlled by one or more tax exempt entities?

To work out if your company is controlled by one or more exempt entities, you will need to consider if one or more exempt entities, their affiliates or both have either:

- shares and other equity interests in your company that give them and/or their affiliates at least 50% of the voting power in your company
- the right to receive at least 50% of any income or capital your company distributes.

- ○ Yes
- ☑ No

Does the company have an Ultimate Holding Company?

A company is an Ultimate Holding Company if it has majority ownership of or controlling interests in the other companies in the consolidated or multiple entry consolidated group. The ultimate holding company may be incorporated in a country other than Australia. More information can be found on the ASIC website and the Corporations Act 2001 where the term 'ultimate holding company' is defined.

- ☑ Yes
- ○ No

What country was the Ultimate Holding Company incorporated in?

AUSTRALIA

What is the Ultimate Holding Company's ABN or ACN?

Company name

CAPIX PTY LTD

Australian Business Number (ABN)

95087364805

Australian Company Number (ACN)

087364805

Registration Date

30/04/1999

Is the company Indigenous owned (where at least 51% of the organisation's members or proprietors are Indigenous)?

- ○ Yes
- ☑ No
- ○ Prefer not to answer

Is the company Indigenous controlled (where at least 51% of the organisation's board or management committee are Indigenous)?

- ○ Yes
- ☑ No
- ○ Prefer not to answer

Which industry does the company mostly operate in?

ANZSIC Division

Select the Australian and New Zealand Standard Industrial Classification (ANZSIC) division that best describes the main business activity of the company.

M - PROFESSIONAL, SCIENTIFIC AND TECHNICAL SERVICES

ANZSIC Class

7000 Computer System Design and Related Services

Contact details

For help with completing this section, please refer to the **Application Guidance**.

Please note that all contacts listed will receive correspondence about the application. Any contact listed may be contacted by the R&DTI Program to provide further information.

Primary company contact details

At least one company contact must be provided.

Title (optional)

Mr

First name

Peter

Last name

Cooney

Position or role

Director

Phone number

For phone numbers outside of Australia, please include the international code (e.g. +64 X XXXX XXXX).

+61419337875

Email

To ensure the integrity of your information, please provide a personal email address. Do not use a generic email address. Using a generic email address may result in correspondence not being received.

peter@capix.net

Main business address

This is the main address where the company does business in Australia.

L 2 696 Bourke St, MELBOURNE VIC 3000

Website (optional)

www.capix.net

Would you like to include an alternate company contact?

- ○ Yes
- ☑ No

Did you rely on advice from a tax agent?

- ○ Yes
- ☑ No

Did you receive advice from an R&D consultant?

Please include details of the primary consultant who provided advice on your application. Please note, primary consultant details are collected for internal reporting only. The primary consultant will not receive correspondence about the application and will not be contacted by the R&D Tax Incentive Program to provide further information.

- ○ Yes
- ☑ No

Application inclusions

For help with completing this section, please refer to the **Application Guidance**.

This application will include:

Select one or more of the options below.

- ☐ Activities with an advance or overseas finding
- ☐ Expenditure paid via a levy to a Research Service Provider
- ☐ Activities conducted by a Research Service Provider
- ☐ Activities conducted by a Cooperative Research Centre
- ☐ Activities conducted by another research organisation
- ☐ Activities conducted under another collaborative agreement
- ☑ None of the above

Will the company be including activities that are excluded from being a core activity in this application?

- ○ Yes, as supporting activities
- ● No

Employees

For help with completing this section, please refer to the **Application Guidance**.

Total number of employees

This is the total number of employees on the company's payroll at the end of the income period most recently completed (including working directors, partners, proprietors, full time, part time, and casual staff).

For consolidated groups, this will be the total employee numbers for the entire group.

8

Number of employees engaged in R&D (person years)

This is the full time equivalent (FTE) number of staff (including working directors, partners, proprietors, full time, part time, and casual staff) employed by the company on eligible R&D in the most recently completed income year.

4

Number of independent contractors engaged in R&D activities

Enter the total number of independent contractors the Company had on the payroll at the most recent pay period undertaking eligible R&D.

0

Finance

For help with completing this section, please refer to the **Application Guidance**.

What is the company's most recently completed income year?

2024/25

For your most recently completed income period, what was the company's taxable income or loss (across all companies)?

This is the company's taxable income or loss for the most recently completed income year. Losses should be shown as negative figures.

AUD -542,095.00

For your most recently completed income period, what was the company's aggregated turnover?

AUD 734,092.00

For your most recently completed income period, how much revenue did the company earn (across all companies) from export sales?

This is the company's total revenue from export sales for the income year covered by this application as reported in the company's business activity statement provided to the Australian Taxation Office. The total revenue for the entire income year should be included, and this may require a company to add up the individual export sale amounts provided in their periodic business activity statements for the income year.

AUD 654,020.00

For the most recently completed income period, what was the company's R&D Expenditure?

This is the total registered R&D expenditure in the most recent income year. If the actual value is unavailable, please provide an estimate.

AUD 715,088.00

Projects and activities

Project - CAPIX AI Corporate Cash Management (P160E5J1M)

For help with completing this section, please refer to the **Application Guidance**.

How much did the R&D Tax Incentive influence whether you went ahead with this R&D project?

The Department would like to know how important the R&D Tax Incentive program is to your project decision making.

Significantly

Name for this project

If you have registered this (project/activity) before and this extends into the financial year that you're applying for, please use the same name.

CAPIX AI Corporate Cash Management

Project reference description (optional)

This is an optional field to insert your internal reference.

CAPIX AI Corporate Cash Management: Smarter forecasting and liquidity optimization for global treasuries.

What is the expected duration of this project?

Enter the expected start and end dates (month and year) for the project.

May 2023 to Dec 2026

What is the total estimated expenditure for the project?

We expect you to make records around the time you plan and conduct your activities. These are called contemporaneous records. These records will be the strongest evidence to support your claims when we review your application or registration.

AUD 2,000,000.00

What are the objectives of this project?

Enter a maximum of 4000 characters.

The project objectives are:

Core Objective:
Develop proprietary machine learning models (e.g., LSTM, transformer-based) to automate and enhance multi-currency cashflow forecasting accuracy by 50%+ over traditional methods.
To develop new quantitative AI models specifically for cashflow analytics.

Technical Innovation:
Design an AI-driven anomaly detection system using unsupervised learning (e.g., autoencoders) to identify real-time cashflow irregularities and FX risks.

Deployment Goal:
Integrate explainable AI (XAI) techniques to ensure transparency and compliance in autonomous treasury decision-making for global enterprises.
To develop new and unique AI models for the sector that are optimised for our sector.

Describe what documents you have kept, or intend to keep, in relation to the activities in your project.

Please describe the type of documents that (you) have and will be kept to record the details of the project. These records should include details of the experiments to be performed, including the reasons for undertaking the experiment, the results of the experiments and the conclusions drawn.

Enter a maximum of 4000 characters.

Software Design documents.
Software Architecture documents
Project Plan documents.
Test Plan documents.
Software Test Kit documents.
Labour timesheets.
Data collection logs.

Plant and facilities address

Is the location of majority of R&D activities the same as the Main business address provided?

- [x] Yes
- [ ] No

Briefly describe the plant and facilities to be allocated to the project (specialist equipment, facilities etc).

Explain how the particular plant or facility will be used in the conduct of R&D activities.

Enter a maximum of 4000 characters.

All development and facilities for the project are hosted in the Microsoft Azure datacentre in Sydney Australia. We also run specialist computer devices for operating specialist Artificial Intelligence models in our Melbourne office. We have several Linux-based mini-servers for development and running models in our office.

For the selected income period, how much was spent on feedstock inputs?

Enter 0 if there is no spend related to feedstock inputs.

AUD 0.00

Which field of research best describes the majority of R&D activities in this project?

ANZSRC Division

46 Information and Computing Sciences

ANZSRC Group

4602 Artificial intelligence

To establish whether the Company will be a beneficiary of the proposed R&D activities, please describe whether the Company will:

- effectively own the know-how, intellectual property, or other results arising from the project;
- have control over the direction and conduct of the R&D activities; and
- bear the financial burden of carrying out the activities

Enter a maximum of 4000 characters.

The company owns all of the Intellectual Property for the project. There are agreements in place with all staff, contractors and suppliers assign all and any copyrights and patents exclusively to the company.

Core R&D activity - CAPIX AI CashFlow Forecasting — FX Volatility and Seasonality Extensions (P8HXVJ5N6)

You must conduct or plan to conduct, at least one eligible core R&D activity to register for the R&D Tax Incentive.

Section 355-25(1) of the Income Tax Assessment Act 1997, the law that applies to the program, states:

Core R&D activities are experimental activities:

(a) whose outcome cannot be known or determined in advance on the basis of current knowledge, information or experience, but can only be determined by applying a systematic progression of work that:

(i) is based on the principles of established science; and

(ii) proceeds from hypothesis to experiment, observation and evaluation, and leads to logical conclusions; and

(b) that are conducted for the purpose of generating new knowledge (including new knowledge in the form of new or improved materials, products, devices, processes or services)

For further information about core R&D activities read the **R&D Tax Incentive Guide to Interpretation**.

For help with completing this section, please refer to the **Application Guidance**.

Is it your intention to list a previously registered core R&D activity for this application?

- [ ] Yes
- [x] No

Name for this core R&D activity

If you have registered this core R&D activity before please use the same name.

CAPIX AI CashFlow Forecasting — FX Volatility and Seasonality Extensions

Describe the core R&D activity

Briefly describe what will be conducted and what is planned to be achieved.

Enter a maximum of 4000 characters.

This core R&D activity continues and progresses the multi-year experimental work registered under project P160E5J1M (CAPIX AI Corporate Cash Management). It directly extends the FY24 core activity (CAPIX Artificial Intelligence CashFlow Forecasting and Management, registered under prior reference IR2411898), maintaining methodological continuity while addressing two specific limitations identified in the FY24 evaluation: (a) the FY24 hybrid LSTM/Transformer baseline relied on raw FX rates as input features, and (b) the FY24 model showed reduced predictive accuracy on cashflow series characterised by strong, mixed-frequency seasonal patterns.

During the FY25 income period (Jul 2024 – Jun 2025) the activity extends the validated FY24 hybrid LSTM/Transformer/autoencoder forecasting baseline along two specific progression axes:

1. Integration of FX volatility-derived features into the input feature space, including rolling realised volatility, GARCH(1,1) conditional volatility, and learned currency-pair correlation embeddings, replacing or augmenting the raw FX rate inputs used in FY24.

2. Application of NeuralProphet additive seasonality decomposition as a preprocessing layer that decomposes cashflow series into trend and weekly/monthly/quarterly/yearly seasonality components prior to LSTM/Transformer prediction.

The activity is conducted as a systematic progression of work — hypothesis, experimental implementation, observation, and evaluation — using stored treasury data and deployed on the Microsoft Azure South-East datacentre infrastructure. Forecast performance is compared against the FY24 baseline arm using Mean Absolute Percentage Error (MAPE) at 30/60/90-day horizons, directional accuracy, and FX-stressed resilience under simulated rate shocks.

Ablation analysis attributes the marginal contribution of each progression axis (FX volatility features alone, NeuralProphet decomposition alone, and the combined arm), enabling the company to determine experimentally whether each axis contributes independent improvement and whether their effects compound.

The outcome of the activity cannot be known or determined in advance on the basis of current knowledge, information, or experience worldwide. There is no published benchmark for NeuralProphet additive seasonality decomposition feeding into attention-based deep learning predictors on sparse, multi-currency corporate treasury cashflow series. The marginal contribution of FX volatility-derived features over raw FX rate features is similarly untested in this domain; volatility-derived features can either improve forecasts by exposing risk regimes or degrade them by introducing noise correlated with the target series, and the net domain-specific effect is only knowable through empirical experiment. Investigation of academic literature, arXiv preprints, vendor documentation, and prior-art searches confirmed that the proposed combination has no reproducible benchmark in the public domain. Industry treasury vendors publish marketing claims rather than reproducible architecture details.

The activity is planned to achieve measurable improvement in multi-currency cashflow forecast accuracy over the FY24 baseline, particularly for series characterised by strong seasonal patterns or significant FX exposure.
The new knowledge generated includes:
(a) a treasury-specific FX feature library with measured ablation contribution to forecast accuracy and stress resilience;
(b) a reproducible empirical benchmark for NeuralProphet decomposition fed into LSTM/Transformer ensembles in the corporate treasury domain; and
(c) refined data preprocessing and model integration protocols for high-frequency FX inputs into deep learning treasury models.

These outcomes are intended to feed directly into the next FY26 progression of project P160E5J1M, in which agentic-AI components will rely on the FY25 forecasting layer as an explainable foundation for autonomous treasury decision-making.

Which project is this core R&D activity related to?

Select the project that this core R&D activity relates to.

CAPIX AI Corporate Cash Management

Does this activity fall into one of the categories that exclude it from being a core R&D activity?

There are activities that are excluded from being core R&D activities. They cannot be registered as core R&D activities but may be eligible as supporting R&D activities. You must check if any of your activities are excluded from being core R&D activities. The law, at s 355-25(2) of the ITAA 1997 lists those activities that cannot be core R&D activities for the R&D Tax Incentive.

- ○ Yes
- ☑ No

Who was this core R&D activity performed by?

- ☑ Only the R&D Company
- ○ The R&D Company and Others (including RSPs and CRCs)
- ○ Either a subsidiary; by more than one company within the group; or by Others (including RSPs and CRCs) working with at least one other company within the group
- ○ Only Others (including RSPs and CRCs)

Does this core R&D activity commence after the end of your income period for this application?

Where a core R&D activity is planned to occur in a future income year, you will need to provide the title of the core R&D activity, its start and end date, a brief description of the activity, and the new knowledge the activity is intended to create.

- ○ Yes
- ☑ No

Enter the actual or anticipated start and end dates for this core R&D activity

The start and end dates (month and year) for the core R&D activity must fall within the dates specified for the related project.

Jul 2024 to Jun 2025

Is this core R&D activity covered by an Industry Research and Development Determination?

For more information about Determinations, please refer to business.gov.au.

- ○ Yes
- ☑ No

How did the company determine that the outcome could not be known in advance?

Select all options that apply.

- [x] There was no applicable information in scientific, technical, or professional literature or patents
- [ ] Experts in the field provided advice that there wasn't a solution that could be applied
- [x] There wasn’t a way to adapt solutions from other companies in, and out of, Australia
- [ ] Other
- [ ] The company did not look into existing knowledge

Please explain what sources were investigated and what information was found.

Enter a maximum of 4000 characters.

Sources investigated:

1. Academic literature on hybrid deep-learning architectures for financial time-series forecasting, accessed via Google Scholar, arXiv (cs.LG, q-fin.CP, q-fin.ST), IEEE Xplore, and SSRN. Searches covered LSTM, Transformer, and ensemble approaches applied to cashflow and currency forecasting.

2. Specific literature on NeuralProphet (Triebe et al., 2021, extending Facebook Prophet), including official documentation, GitHub issue tracker, and benchmarking studies, with focus on integrations of NeuralProphet decomposition output with downstream deep-learning predictors.

3. Literature on GARCH and stochastic volatility modelling for foreign exchange rates (Engle, Bollerslev) and on the use of volatility-derived features in financial prediction.

4. Industry reports on treasury pain points, multi-currency cashflow forecasting, and FX risk management, including Association for Financial Professionals (AFP) survey data, treasury vendor reports, and major advisory

firm publications.

5. Vendor product documentation and marketing materials from established treasury technology providers (Kyriba, Cashforce, FIS, GTreasury, ION Treasury, Coupa) to identify the architectures and feature sets in commercial use.

6. ERP data structures and integration documentation from SAP, Oracle, and Microsoft Dynamics, focusing on cashflow-relevant data feeds and multi-currency FX transactional structures.

7. Microsoft Azure Machine Learning technical documentation covering model training, deployment, and ablation tooling.

Findings:

- No published hybrid model combines NeuralProphet additive seasonality decomposition with downstream LSTM/Transformer ensemble prediction for multi-currency corporate treasury cashflow. NeuralProphet benchmarks exist for retail demand, energy load, and macroeconomic series, but the specific composition with attention-based downstream predictors on sparse multi-currency treasury cashflow is absent from the public literature.

- The marginal contribution of FX volatility-derived features (realised volatility, GARCH(1,1) conditional volatility, currency-pair correlation embeddings) over raw FX rate inputs is untested in the treasury cashflow domain. Such features are well-studied for asset returns and options pricing but not as inputs to cashflow forecasting models, and the net effect on forecast accuracy is unknowable without empirical experiment.

- Industry treasury vendors publish marketing claims about AI-powered forecasting but do not publish reproducible architectures, ablation results, or methodology details. No commercial product was identified that could be adapted to provide the FX-aware, seasonality-decomposed forecasting capability under investigation.

- Rule-based and conventional AI systems described in industry reports continue to miss a substantial proportion of cashflow anomalies under FX-stressed scenarios, consistent with AFP research and confirmed during the company's FY24 baseline evaluation.

- ERP data structures vary materially between SAP, Oracle, and Dynamics environments. Existing approaches in the literature do not address the integration of high-frequency FX volatility features with heterogeneous ERP transactional inputs.

Novelty — why competent professionals cannot predict the outcome:

1. The interaction between NeuralProphet decomposition and attention-based downstream predictors on sparse multi-currency treasury cashflow has no published benchmark; the outcome of the proposed combination cannot be inferred from any existing reference.

2. The marginal contribution of FX volatility-derived features over raw FX rate features is domain-specific and untested in this context. The same class of feature can improve or degrade forecasts depending on predictor architecture, target series characteristics, and FX exposure profile, making the net effect knowable only through systematic experiment and ablation.

Why couldn’t a competent professional have known or determined the outcome in advance?

Enter a maximum of 4000 characters.

A competent professional in machine learning, quantitative finance, or treasury technology — even with deep expertise in each technique — could not have determined this activity's outcome in advance. Four specific knowledge gaps prevent prediction:

1. No published benchmark for the architectural combination. The work combines NeuralProphet additive seasonality decomposition (Triebe et al., 2021) as a preprocessing layer with a downstream LSTM/Transformer ensemble for sparse, multi-currency corporate treasury cashflow. Each component has independent literature — NeuralProphet for retail/energy/macro forecasting, attention predictors for asset returns — but no published study

combines them on this class of target. Deep-learning architectures exhibit emergent behaviour from feature-target-architecture interactions not predictable from component-wise analysis.

2. Untested marginal contribution of FX volatility features. The contribution of FX volatility-derived features (realised volatility, GARCH(1,1) conditional volatility, currency-pair correlation embeddings) over raw FX rates is untested for treasury cashflow. The same feature class can improve forecasts by exposing risk regimes or degrade them by introducing target-correlated noise; the dominant effect depends on architecture, target profile, and FX regime, and cannot be determined without empirical test.

3. Proprietary commercial products do not reveal outcomes. Treasury vendors (Kyriba, Cashforce, FIS, GTreasury, ION) publish marketing claims but not reproducible architectures, ablation results, or quantitative comparisons.

4. Domain-specific datasets are not in the public domain. Forecasting performance is dominated by the target organisation's transactional patterns (payroll cycles, supplier terms, FX exposure profile, sector seasonality), which are not in public datasets and do not generalise across organisations.

Outcome is therefore knowable only by applying a systematic progression of work — hypothesis, experiment, observation, evaluation — as conducted in this core activity.

What is the hypothesis?

Enter a maximum of 4000 characters.

The core R&D activity proceeds under a structured hypothesis that extends the validated FY24 baseline along two specific progression axes. It comprises a base claim (FY24 architecture as control) and three progression claims specific to FY25.

BASE — FY24 architecture as control:

The FY24 core activity (registered under IR2411898) hypothesised that an AI-driven hybrid LSTM/Transformer/autoencoder ensemble would significantly outperform traditional rule-based cashflow forecasting on multi-currency corporate treasury data. This base architecture remains operational through FY25 and provides the experimental control against which FY25 progression is measured.

AXIS 1 — FX volatility-derived features:

We hypothesise that augmenting the input feature space of the hybrid predictor with FX volatility-derived features — (a) rolling realised volatility for relevant currency pairs, (b) GARCH(1,1) conditional volatility estimates, and (c) learned currency-pair correlation embeddings — will produce a measurable reduction in Mean Absolute Percentage Error (MAPE) at 30/60/90-day horizons, particularly under FX-stressed conditions. The expected mechanism is that volatility-derived features expose risk-regime information that raw FX rates do not, allowing the predictor to condition forecasts on the prevailing volatility regime. Whether this overcomes the alternative risk that volatility features contribute target-correlated noise is the experimental question.

AXIS 2 — NeuralProphet seasonality decomposition:

We hypothesise that decomposing target cashflow series with NeuralProphet's additive seasonality components (trend plus weekly, monthly, quarterly, yearly cycles) prior to LSTM/Transformer prediction will produce a measurable reduction in MAPE on cashflow series with strong, mixed-frequency seasonal patterns — payroll cycles, end-of-month settlements, quarterly tax payments — for which the FY24 hybrid demonstrated reduced accuracy. The expected mechanism is that explicit decomposition reduces modelling burden on the downstream attention-based predictor, freeing capacity for the residual non-seasonal signal.

COMBINED HYPOTHESIS:

We further hypothesise that the two progression axes contribute independent and additive improvements over the FY24 baseline, such that a combined arm (FX volatility features + NeuralProphet decomposition + LSTM/Transformer/autoencoder ensemble) produces a reduction in 30/60/90-day MAPE of at least 10 percent over the FY24 baseline on FX-exposed multi-currency series, and maintains this advantage under simulated FX shocks where the FY24 baseline materially degrades.

REJECTION CRITERION:

The hypothesis is rejected if, after the FY25 progression of work, neither axis produces a statistically distinguishable improvement over the FY24 baseline on the company's stored treasury cashflow data, evaluated using MAPE at 30/60/90-day horizons, directional accuracy, and FX-stressed resilience.

What is the experiment and how will it / did it test the hypothesis?

Enter a maximum of 4000 characters.

The experiment is a phased, controlled test of the FY25 progression hypothesis, conducted over the income period (Jul 2024 – Jun 2025) on the company's stored corporate treasury cashflow dataset. It is designed to attribute marginal contribution to each progression axis (FX volatility features, NeuralProphet seasonality decomposition) and to the combined arm.

EXPERIMENTAL DESIGN — phased deployment with persistent FY24 baseline control:

Phase 0 — FY24 baseline (control). The hybrid LSTM/Transformer/autoencoder ensemble registered as P5D5WJ0S0 in FY24 continues to operate on the same data feeds during FY25; its forecasts are the experimental control against which all FY25 arms are measured.

Phase 1 — Infrastructure migration (completed 28 Sep 2024). Forecasting workloads migrated to the Microsoft Azure South-East datacentre and Azure ML compute, providing parallel-arm capacity, ablation tooling, and FX-data ingestion bandwidth.

Phase 2 — FX volatility features (from 11 Feb 2025). A feature engineering pipeline was added computing rolling realised volatility, GARCH(1,1) conditional volatility, and currency-pair correlation embeddings from high-frequency FX rate feeds. The augmented arm runs in parallel with the FY24 baseline against the same target cashflow data.

Phase 3 — NeuralProphet seasonality preprocessor (from 22 Mar 2025). A NeuralProphet preprocessing layer decomposes target cashflow series into trend and weekly/monthly/quarterly/yearly seasonality components, passed as auxiliary features to the LSTM/Transformer ensemble. Runs in parallel with the FY24 baseline and Phase 2.

Phase 4 — Combined arm and ablation (Apr–Jun 2025). The combined arm (FX volatility + NeuralProphet + LSTM/Transformer/autoencoder) runs alongside the baseline and single-axis arms, enabling ablation comparison of marginal contribution per axis and any non-additive interaction.

HOW THE EXPERIMENT TESTS THE HYPOTHESIS

- Phase 2 vs FY24 baseline tests Axis 1 (FX volatility features improve accuracy).
- Phase 3 vs FY24 baseline tests Axis 2 (NeuralProphet decomposition improves seasonal-series accuracy).
- Phase 4 vs single-axis arms tests whether axes contribute independent and additive improvements or interact non-additively.
- All arms are stress-tested by injecting simulated FX rate shocks calibrated to FY25 historical volatility surfaces; resilience scores are compared across arms.

METRICS

MAPE at 30/60/90-day horizons; directional accuracy; statistical comparison of arm-versus-baseline differences. Stress resilience score (MAPE retention under simulated FX shocks). Anomaly detection precision/recall continuing from FY24 with FX-driven anomalies as a new sub-category.

Conclusions reached within the income period are reported in the Conclusions field. Further validation continuing into FY26 will be reported in the FY26 application.

How did you evaluate or plan to evaluate results from your experiment?

Enter a maximum of 4000 characters.

Evaluation of the experiment is structured across four interlocking dimensions, each tied to a hypothesis claim and measurable on the company's stored treasury cashflow data.

1. QUANTITATIVE FORECAST ACCURACY

Primary metric: Mean Absolute Percentage Error (MAPE) at 30/60/90-day horizons, computed arm-by-arm against actual realised cashflow. Each FY25 arm is compared against the persistent FY24 baseline (Phase 0). Secondary metrics: directional accuracy (proportion of forecasts correctly anticipating cash surplus or deficit); marginal contribution per progression axis (delta MAPE attributable to FX volatility features alone, NeuralProphet alone, and combined). Statistical comparison against the baseline uses appropriate methods for paired forecast series, with confidence intervals reported for each marginal contribution estimate.

2. FX-STRESSED RESILIENCE TESTING

Simulated FX rate shocks are injected into the input data — sudden EUR/USD, GBP/USD, and JPY/USD movements calibrated to FY25 historical volatility surfaces. Each arm's forecast is recomputed under stress and a resilience score is captured (proportion of forecasts retaining accuracy below a defined MAPE threshold). The hypothesis predicts volatility-aware arms maintain accuracy under stress where the FY24 baseline degrades; the resilience score directly tests this.

3. EXPLAINABILITY AND FEATURE-IMPORTANCE ANALYSIS

SHAP and LIME analyses are applied to the FX volatility features and NeuralProphet decomposition outputs to determine whether observed forecast improvement is attributable to the new features rather than spurious correlation or data leakage. Feature-importance attribution also informs operator transparency — treasury team override rates and trust thresholds are tracked, continuing the FY24 human-AI collaboration evaluation. Auditor feedback on FX-driven anomaly alerts is recorded against compliance standards.

4. CONTINUOUS LEARNING AND DRIFT VALIDATION

Model decay is tracked in the volatile FY25 FX environment. Delta accuracy gain (change in MAPE as models ingest new data through the income period) recorded to test whether the progression-axis features remain stable under non-stationary conditions, or introduce concept drift not exhibited by the FY24 baseline.

WHY PRE-EXPERIMENT COULD NOT PREDICT RESULTS

The marginal contribution of FX volatility-derived features with attention-based predictors is not knowable without empirical ablation; no published benchmark exists for NeuralProphet decomposition feeding LSTM/Transformer ensembles for treasury cashflow; whether the two axes interact additively is determined by feature-data dependencies measurable only empirically.

Conclusions reached within the income period are reported in the Conclusions field; full statistical validation continuing into FY26 will be reported in the FY26 application.

If you reached conclusions from your experiments in the selected income period, describe those conclusions.

Enter a maximum of 4000 characters.

The following conclusions were reached within the FY25 income period (Jul 2024 – Jun 2025):

1. CONTINUITY OF FY24 BASELINE. The FY24 hybrid LSTM/Transformer/autoencoder ensemble continued in operation throughout FY25 as the experimental control. Forecasting was not interrupted by the Azure South-East migration in September 2024, providing a continuous comparison series against which FY25 arms were measured.

2. INFRASTRUCTURE READINESS (Phase 1, Sep 2024). Migration to the Microsoft Azure South-East datacentre and Azure ML compute provided the parallel-arm capacity, ablation tooling, and high-frequency FX-data ingestion

bandwidth required for the multi-arm experiment. The infrastructure was validated in production by successful deployment of Phases 2 and 3.

3. AXIS 1 — FX VOLATILITY FEATURES (Phase 2, deployed Feb 2025). Following introduction of FX volatility-derived features (rolling realised volatility, GARCH(1,1) conditional volatility, currency-pair correlation embeddings), the augmented arm produced directional improvement in cashflow forecast accuracy over the FY24 baseline on multi-currency series with material FX exposure. The improvement was most evident under simulated FX-stress conditions, consistent with the hypothesised mechanism that volatility-derived features expose risk-regime information absent from raw FX rates. Full statistical validation across the seasonal cycle continues into FY26.

4. AXIS 2 — NEURALPROPHET DECOMPOSITION (Phase 3, deployed Mar 2025). Following integration of the NeuralProphet preprocessing layer, the decomposed-input arm produced directional improvement on cashflow series characterised by strong, mixed-frequency seasonal patterns — payroll cycles, end-of-month settlements, quarterly tax payments — series where the FY24 hybrid had previously demonstrated reduced accuracy. Marginal contribution attribution against Axis 1 continues into FY26.

5. EXPLAINABILITY. Feature-importance analysis within the income period indicated that observed accuracy improvements are attributable to the new FX
volatility and seasonality features rather than to spurious correlation, supporting the inference that the progression axes contribute genuine new predictive information.

6. SCOPE OF FY25 KNOWLEDGE. The income-period evidence supports the hypothesis directionally on each progression axis individually. The combined arm (Phase 4) was deployed in Q4 of the income period (Apr–Jun 2025); the bulk of statistical validation, marginal-contribution attribution, and drift testing continues into FY26 and will be reported in the FY26 application.

The FY25 progression has produced empirical evidence on a research question — the value of FX volatility-derived features and NeuralProphet decomposition in attention-based deep learning treasury cashflow forecasting — for which no public benchmark previously existed.

What evidence did the company keep about this core R&D activity?

Select all that apply.

- [x] Evidence of your hypothesis and design of your experiments
- [x] Documented results and evaluation of your experiments
- [x] Evidence of revisions to your experiment in response to previous experimental results
- [x] Evidence of searches or enquiries you made to find current knowledge
- [x] Evidence to show that you could only determine the outcome of the core R&D activity by conducting experiments as part of a systematic progression of work
- [ ] Other
- [ ] The company did not keep records

Did you conduct this core R&D activity for a purpose of generating new knowledge?

To be an eligible core R&D activity, one of your purposes to conduct R&D needs to be to generate new knowledge.

- (•) Yes
- ( ) No

What new knowledge was / will this core R&D activity (be) intended to produce?

Enter a maximum of 4000 characters.

Your description should include sufficient and relevant detail so that the Department can understand the new knowledge the core R&D activity is intended to generate.

The core R&D activity is intended to produce new knowledge across five distinct dimensions, none currently available in the public scientific or technical literature, and none obtainable without the experimental work registered as this activity.

1. EMPIRICAL CHARACTERISATION OF FX VOLATILITY FEATURES IN TREASURY CASHFLOW. The activity will produce

the first empirical characterisation of the marginal contribution of FX volatility-derived features (rolling realised volatility, GARCH(1,1) conditional volatility, currency-pair correlation embeddings) over raw FX rate inputs in deep learning models for multi-currency treasury cashflow forecasting. This includes ablation contribution per feature class and resilience under simulated FX stress. Such features are well-studied for asset return prediction and options pricing but not for cashflow forecasting.

2. REPRODUCIBLE BENCHMARK FOR NEURALPROPHET FEEDING ATTENTION PREDICTORS. The activity will produce a reproducible empirical benchmark for NeuralProphet additive seasonality decomposition (Triebe et al., 2021) feeding LSTM/Transformer ensemble predictors for sparse multi-currency treasury cashflow. NeuralProphet has published benchmarks in retail demand, energy load, and macroeconomic forecasting; the activity extends the benchmark space to a previously uncharted domain.

3. PREPROCESSING-PIPELINE INTERACTION EFFECTS. The activity will produce empirical evidence on how the two progression axes interact when combined. Whether they contribute additively, sub-additively, or super-additively is determined by data-flow dependencies not predictable from component-wise analysis. Ablation will produce specific knowledge of these interaction effects for the treasury cashflow domain.

4. ERP–FX INTEGRATION PROTOCOLS. The activity will produce refined data-preprocessing protocols for integrating high-frequency FX feature streams with heterogeneous ERP transactional data (mixed structured and unstructured inputs from SAP, Oracle, and Microsoft Dynamics). Existing literature treats FX feature engineering and ERP data integration in isolation; the activity contributes operational knowledge of their joint preprocessing.

5. EXPLAINABILITY-INFORMED FEATURE VALIDATION FOR TREASURY-DOMAIN ML. The activity will contribute knowledge of how feature-importance attribution techniques apply specifically to FX volatility-derived features and NeuralProphet decomposition outputs in the treasury cashflow context, including how attribution informs operator transparency, treasury team trust thresholds, and audit compliance for FX-driven anomaly detection.

PURPOSE OF GENERATING NEW KNOWLEDGE

The new knowledge is the substantive purpose of the activity, not incidental to product development. The progression-axis design was selected to address research gaps identified during FY24 evaluation — gaps for which no commercial product, academic publication, or industry report provided a usable answer.

The experimental design (multi-arm, ablation-based, hypothesis-falsifiable) reflects this new-knowledge purpose: routine product development would not require ablation, a persistent FY24 baseline as control, or simulated-stress resilience testing. The systematic progression of work is conducted for the s355-25(1)(b) purpose of generating new knowledge in the form of improved processes (treasury cashflow forecasting methodology) and services (the company's AI-driven treasury management product).

For your selected income period, what was the estimated expenditure for this core R&D activity?

Enter a reasonable estimate of the expenditure on this core R&D activity for the income year of registration. This should include expenditure on the activity conducted by the company, and contracted expenditure to Research Service Providers or Cooperative Research Centres (if any).

AUD 715,088.00

What is the total actual and reasonably anticipated expenditure for the applicant entity on this activity (per income year)?

Income year 1 is the application income year.

We understand that final accounts may not be available; however, if this is the case, we expect the expenditure amounts you advise to be a reasonable estimate of the figures which will be included in your tax return. Expenditure amounts you expect to incur in the future should be based on reasonable estimates and assumptions.

Expenditure incurred prior to 2024/25

AUD 0.00

Anticipated expenditure income year 2024/25

AUD 715,088.00

Anticipated expenditure income year 2025/26

AUD 0.00

Anticipated expenditure income year 2026/27
AUD 0.00

Anticipated expenditure post 2026/27
AUD 0.00

Total expenditure
AUD 715,088.00

Declare and submit

For help with completing this section, please refer to the **Application Guidance**.

Is the Declarant the same as the Primary Company contact?
The Declarant:

- must be authorised to represent, act on behalf of and bind the company
- must have consented to their personal information being used and disclosed as set out in this form
- may receive correspondence about the application and may be contacted by the Department or the ATO to provide further information.

- [x] Yes
- [ ] No

Declarant details

The declarant details need to be completed and saved before you can submit your application.

Title (optional)
Mr

First name
Peter

Last name
Cooney

Position or role
Director

Phone number
+61419337875

Email
peter@capix.net

Privacy collection statement

The Department of Industry, Science and Resources (the Department) collects, uses and discloses your personal information directly from you, your organisation, and/or your organisation's employees for the purpose of carrying out functions under the Industry Research and Development Act 1986 (IR&D Act) and generally administering the Research and Development Tax Incentive (R&D Tax Incentive). This includes for contacting your organisation, as well as registering your organisation, assessing the eligibility of R&D activities and making findings.

Information may be collected from, or disclosed to, third parties including the ATO and other Australian government entities, as well as officers in other areas of the Department, and used to administer the R&D Tax Incentive, for evaluating and improving the administration of the R&D Tax Incentive, informing policy development and decision-making, as well as to contact organisations to notify them of research services available under the IR&D Act or about other similar programs or services.

The personal information collected by the Department for these purposes includes names, positions, qualifications, office telephone numbers, mobile numbers, and email addresses.

Without this information, we may be unable to process your application for registration.

For more information about the Department's privacy practices, including how to access or correct your personal information or make a complaint, see our privacy policy.

Declaration and submit application

I declare that:

• I, as the declarant, have the authorisation to represent, act on behalf of and bind the company, including to lodge this application;

• all contact persons specified in this form have the necessary authority to represent and act on behalf of the company in relation to the R&D Tax Incentive;

• I consent to my personal information, and I have obtained the consent of all individuals whose personal information is in this form to their personal information, being used for the purposes specified in this form. I will provide each individual with a copy of the privacy collection statement;

• to the best of my knowledge and belief the information in this application is true and correct and accurate in all material details, and that the activities and corresponding expenditure described in this application meet all prescribed eligibility requirements for the R&D Tax Incentive. I understand that giving false or misleading information is a serious offence;

• the R&D entity, while undertaking the activities described in this application, has maintained records, while the activities were conducted, that substantiate the conducting of the activities to be registered for the R&D Tax Incentive; and

• the R&D entity will provide further information as requested by the Department or Innovation Science Australia to support the application for registration in the future, and the company will do so in a reasonable amount of time after receiving a request.

I acknowledge that:

• I have read the R&D Tax Incentive self-assessment tool on the website and determined that our activities are eligible;

• The information provided in this form, and during the course of claiming the R&D Tax Incentive, is protected information under the IR&D Act, and will only be disclosed by the Department for the purposes of registering, assessing and otherwise administering the R&D Tax Incentive (including as described in this form), and as otherwise authorised or permitted by law; and

• it is an offence (subject to a civil penalty) for a person to provide a service that is a 'tax agent service', where that person is not a registered tax agent (refer section 50-5 of Tax Agent Services Act 2009), other than where the service is a legal service in some circumstances.

☑ **I agree**

Potential risks

The following issues have been identified for your application. Please review the following guidance and address any issues as required. You can submit your application by acknowledging that you have considered the guidance to ensure you have correctly assessed your claim as eligible.

I acknowledge I am aware of the potential risks

There are Tax Payer alerts and / or specific guidance relevant to your company's primary industry of operation. Please confirm that you have considered the following guidance to ensure you have correctly assessed your claim as eligible.

ANZSIC Division
M - PROFESSIONAL, SCIENTIFIC AND TECHNICAL SERVICES

ANZSIC Class
7000 Computer System Design and Related Services

Before continuing, please consider the software development sector guide for the R&D Tax Incentive relevant to the ANZSIC Division/Class selected for the industry the company mostly operates in.
Before continuing, please consider the tax payer alert for claiming the R&D Tax Incentive for software development activities relevant to the ANZSIC Division/Class selected for the industry the company mostly operates in.
Before continuing, please consider the tax payer alert for claiming the R&D Tax Incentive for software development activities - Addendum relevant to the ANZSIC Division/Class selected for the industry the company mostly operates in.

☑ I acknowledge that I have reviewed and understood the Tax Payer alerts and / or BGA guidance that are relevant to my company's primary industry of operation.

Acknowledged by:

Name:	Peter michael Cooney
Employer ABN:	81072587976